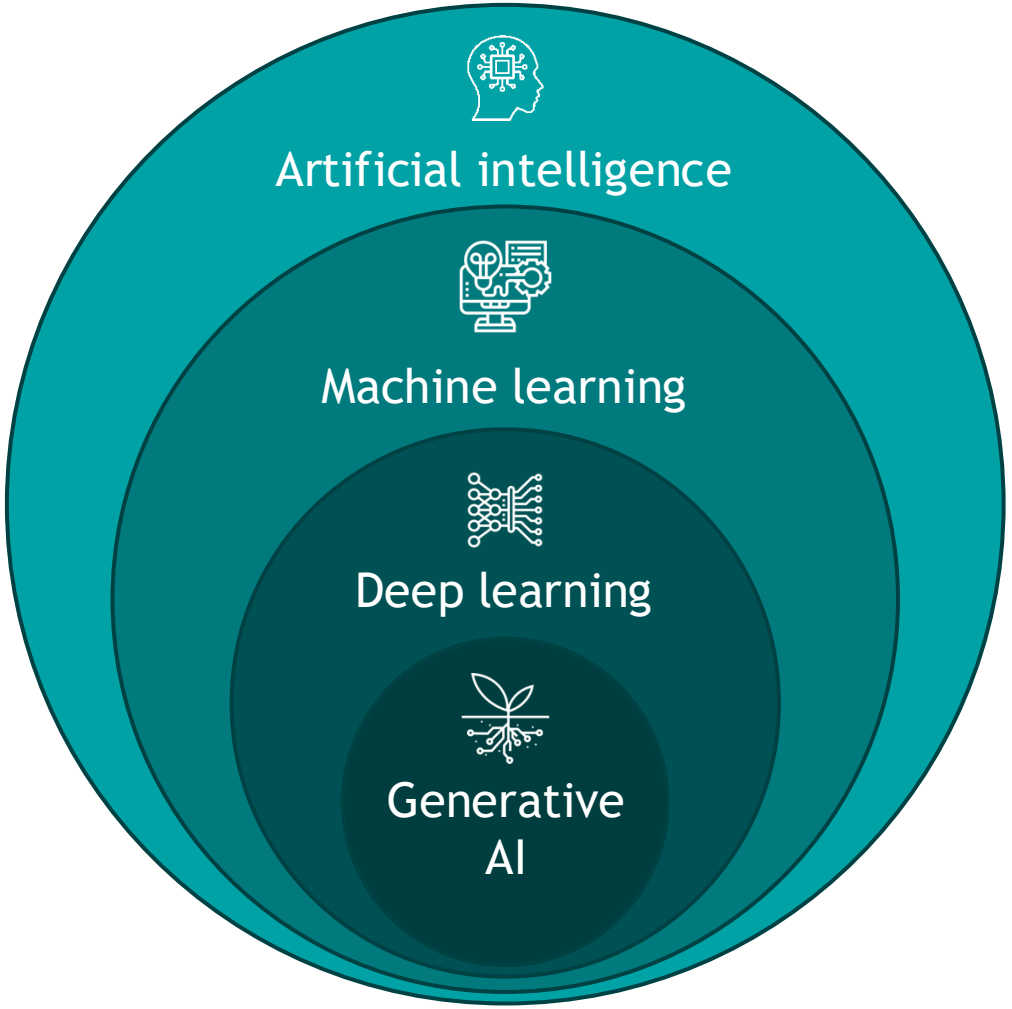
01 Introduction to Neural Networks

Why deep learning?

Building blocks of neural networks

Network architecture

Training neural networks

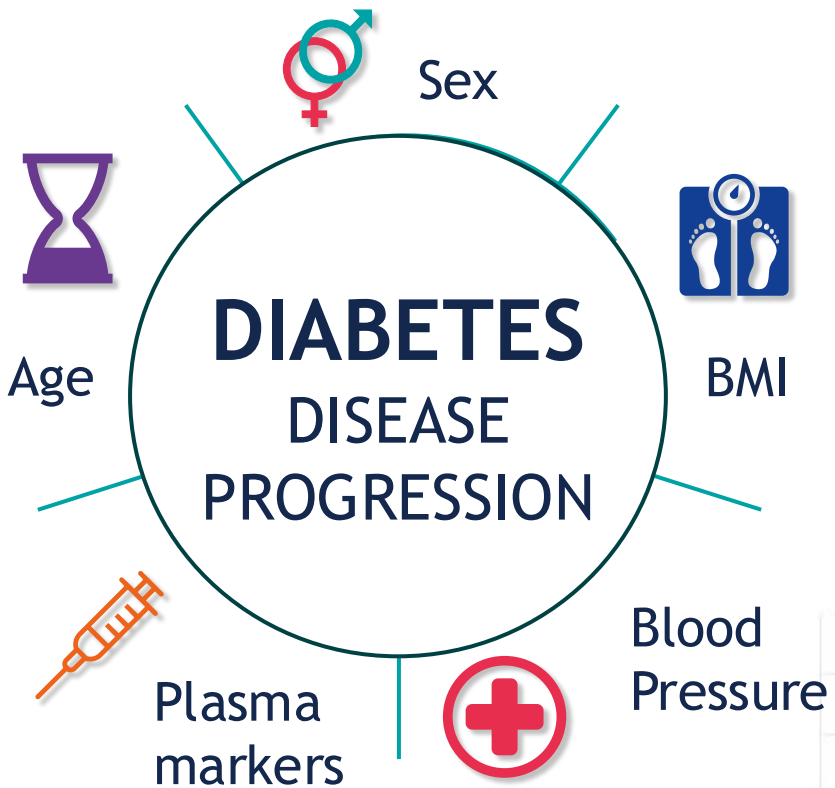


Machine learning – system learns the rules from previous data

Deep learning – machine learning with deep neural networks

→ Same principles as machine learning, but with a specific model architecture

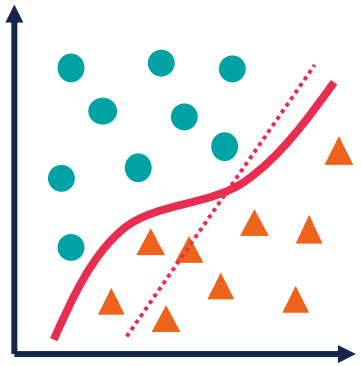
VIB



Feature engineering

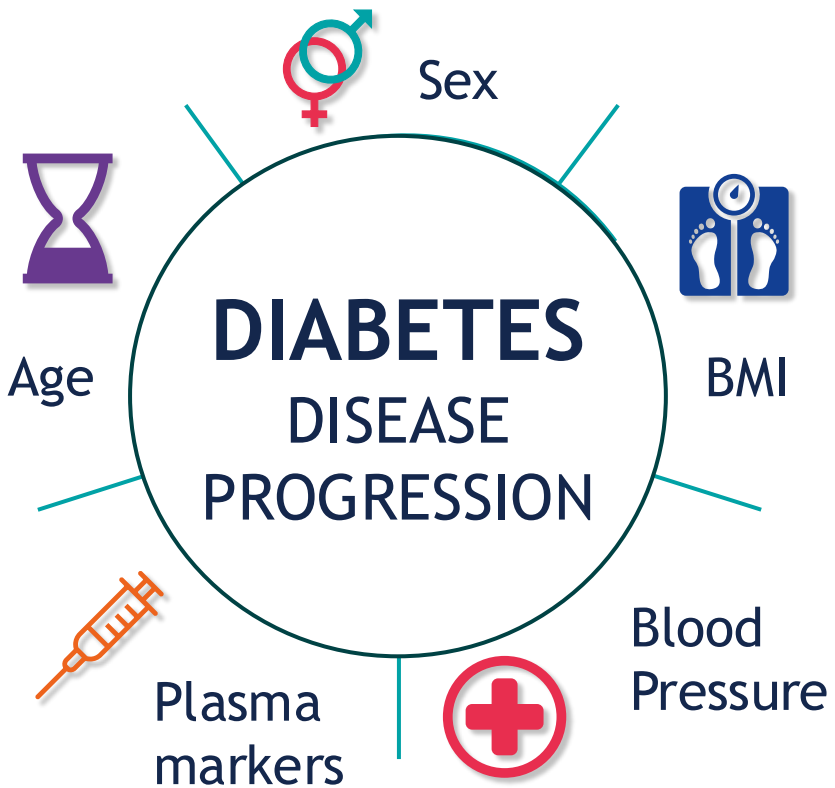
= transforming raw data into meaningful input variables (features)

ML model



ECG features

- Requires domain knowledge (assumptions!)
- Complex/invisible patterns?



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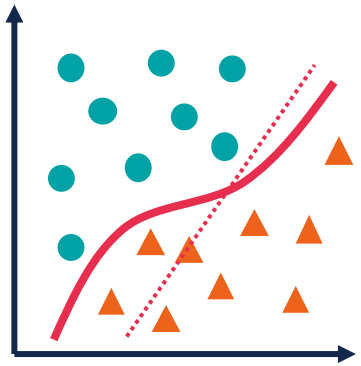
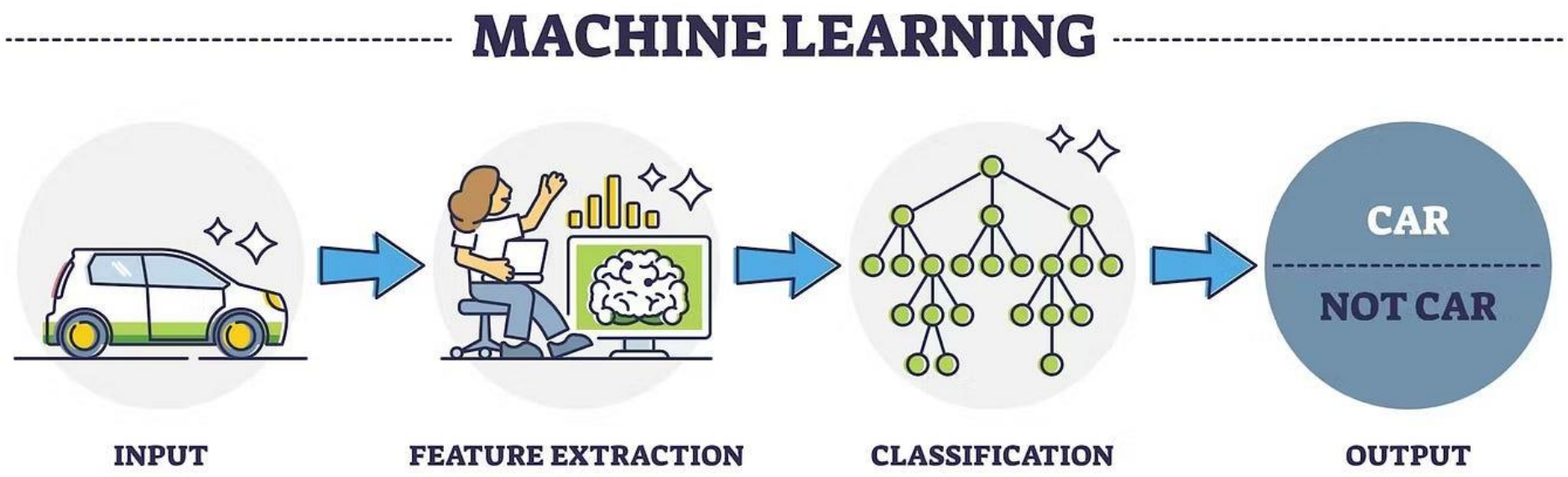


CT features
(e.g. fat distribution)

- Too difficult to compute
- Too expensive to label

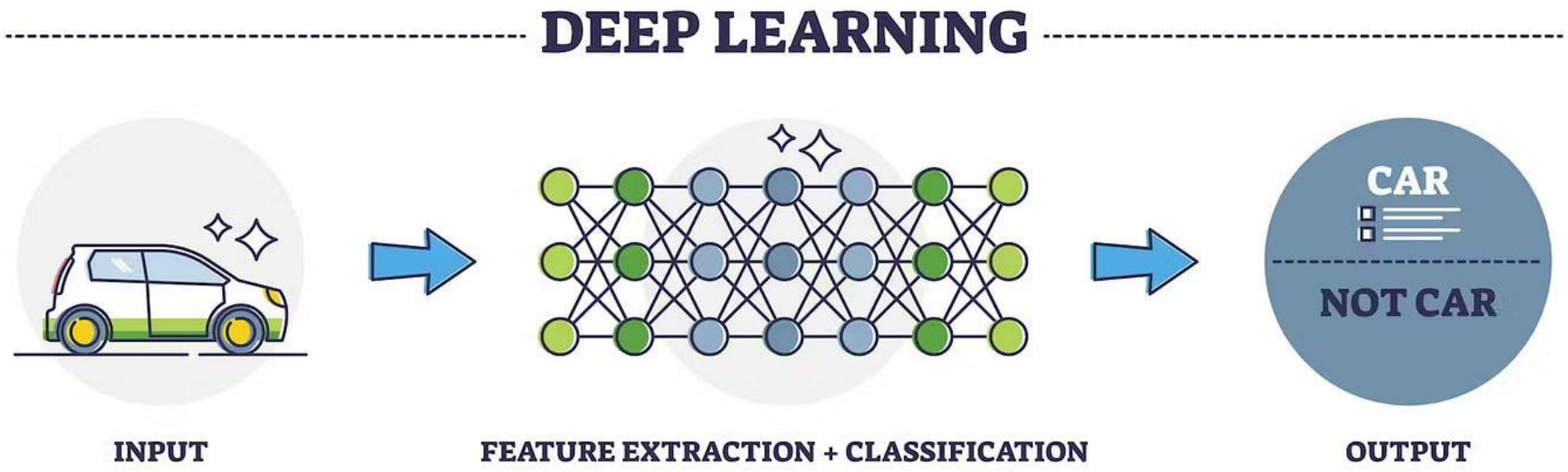
VIB



You decide what information the model gets

- Requires feature engineering
- Can work with thousands of examples
- High interpretability
- Often best choice for tabular data

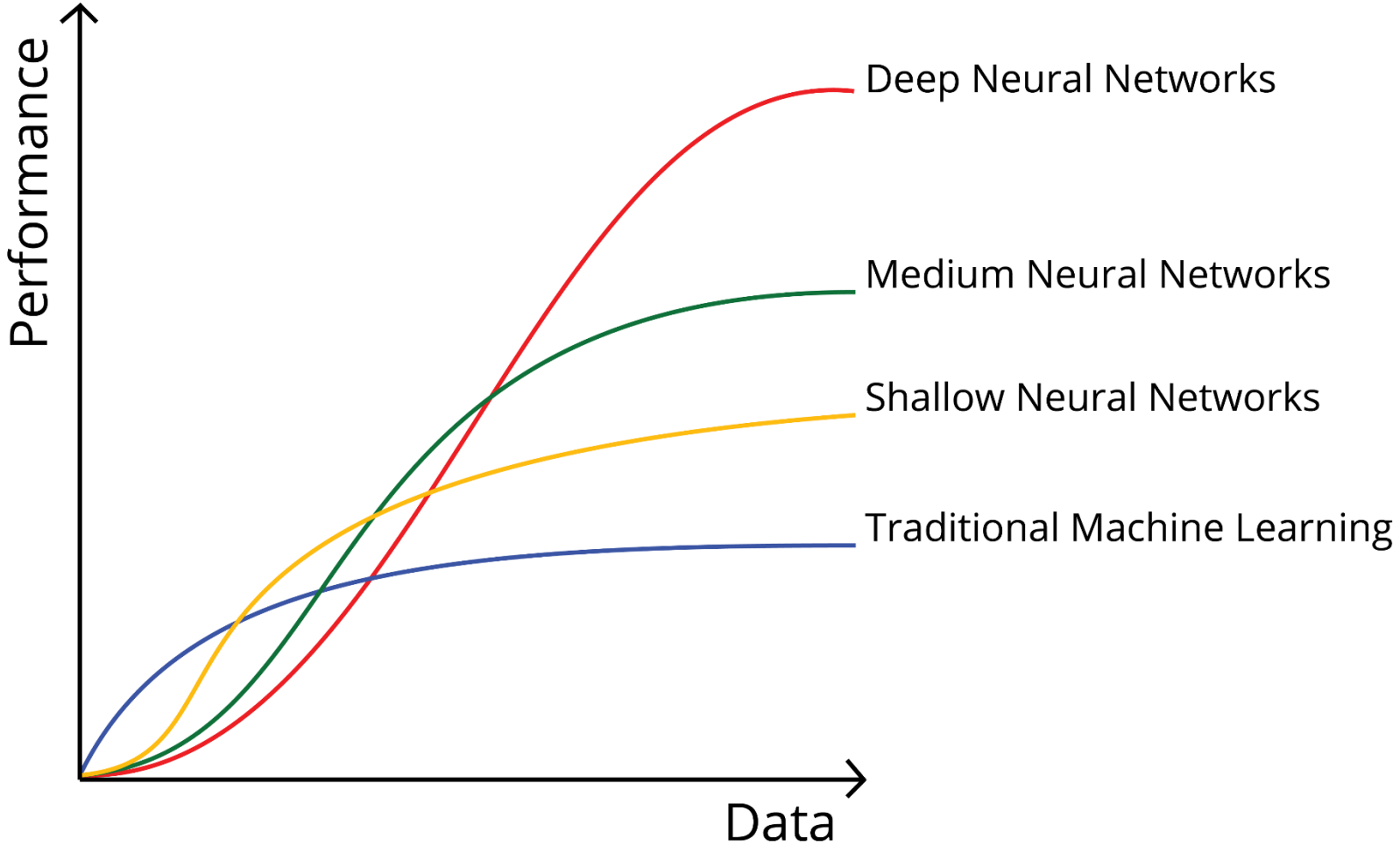
Image adapted from: https://medium.com/@jainpalak9509/why-deep-learning-over-traditional-ml-algorithms-efae1660f3b0



Neural networks learn their own features

- (often) needs millions of examples
- High training cost
- Low interpretability
- Best choice for raw/unstructured data

Image adapted from: https://medium.com/@jainpalak9509/why-deep-learning-over-traditional-ml-algorithms-efae1660f3b0


Performance
Deep Neural Networks
Medium Neural Networks
Shallow Neural Networks
Traditional Machine Learning
Data

Why deep learning?

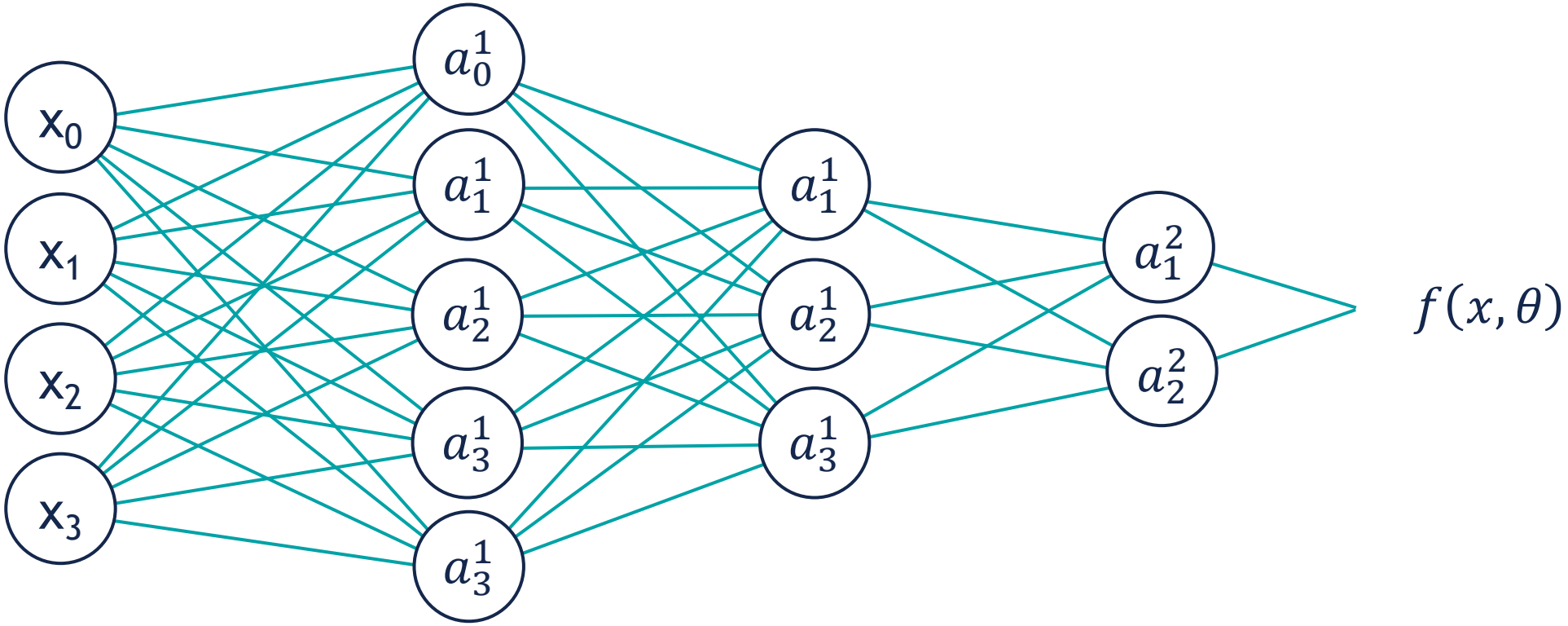
Building blocks of neural networks

Network architecture

Training neural networks

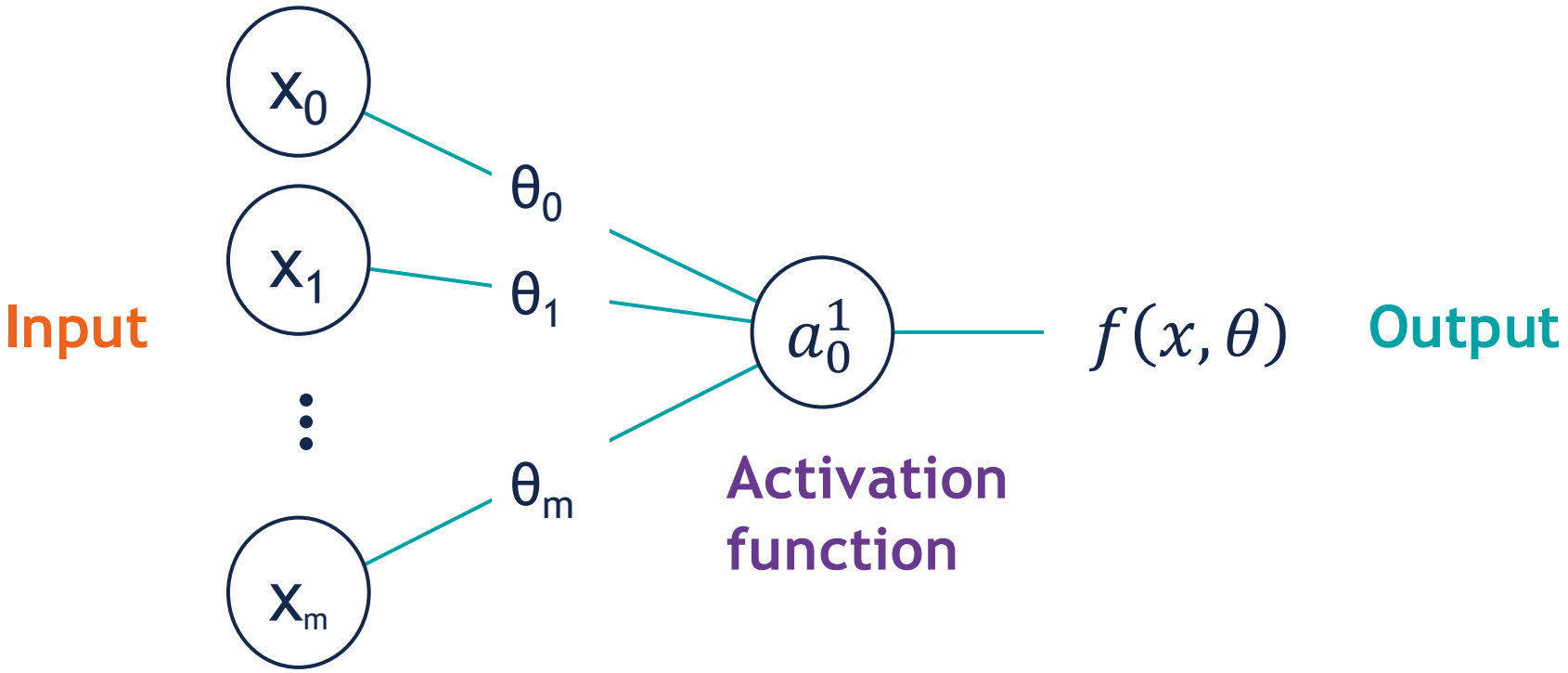
Neural network

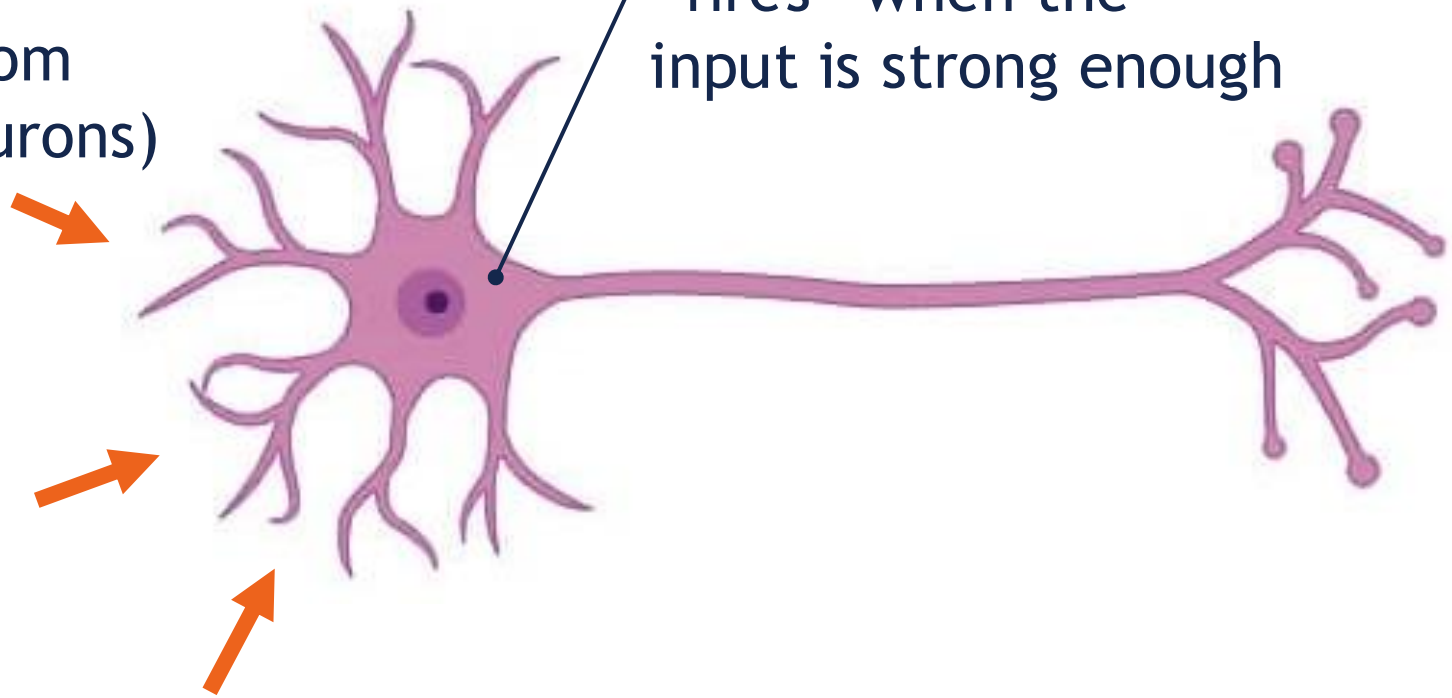
= an architecture that consists of many perceptrons



Perceptron

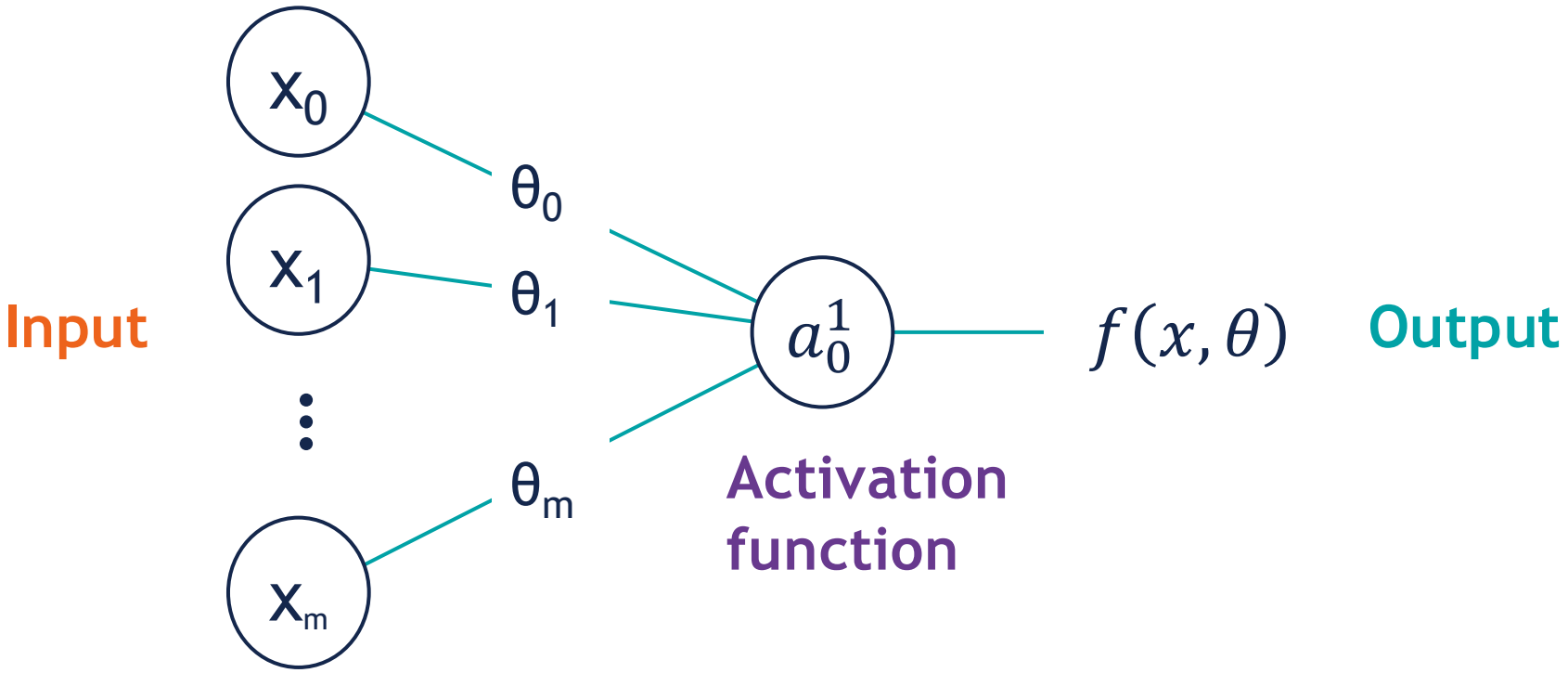
= building block of a neural network




Cell body
“fires” when the input is strong enough
Input (from other neurons)
Output (to other neurons)

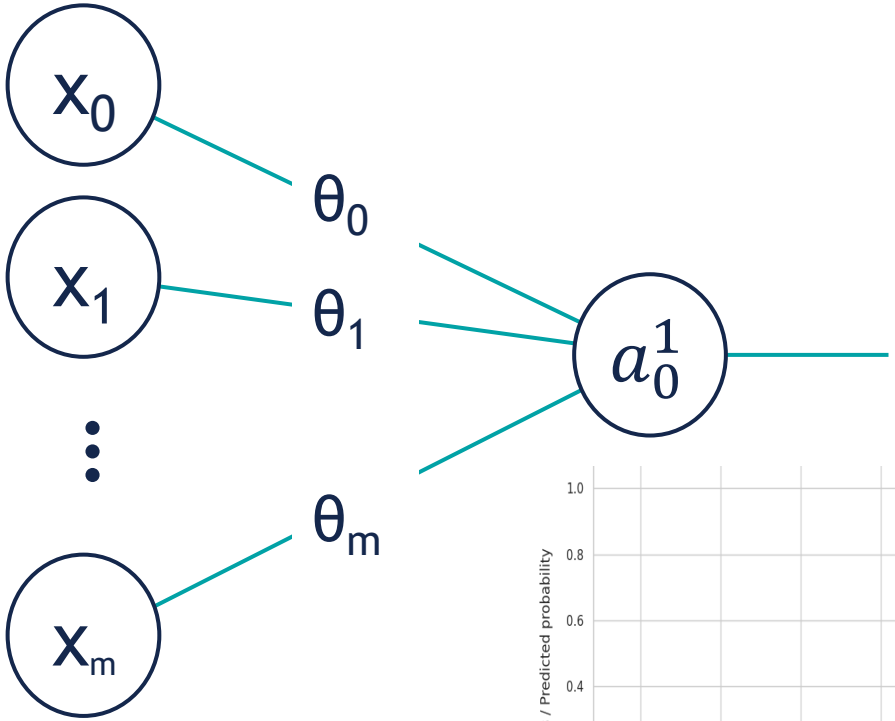
Perceptron

= building block of a neural network

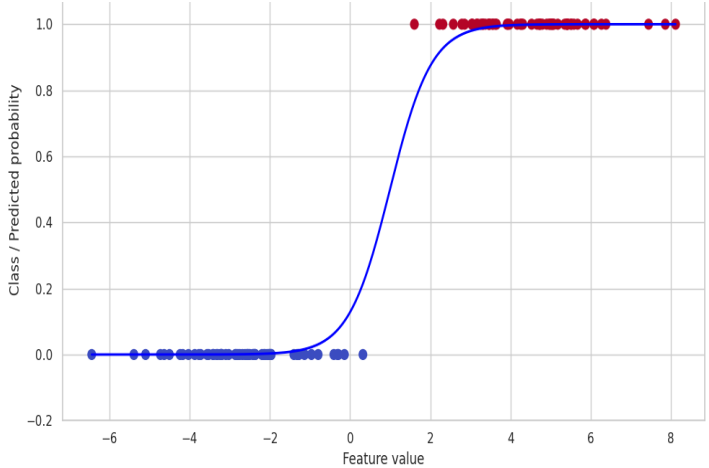


Perceptron

= building block of a neural network



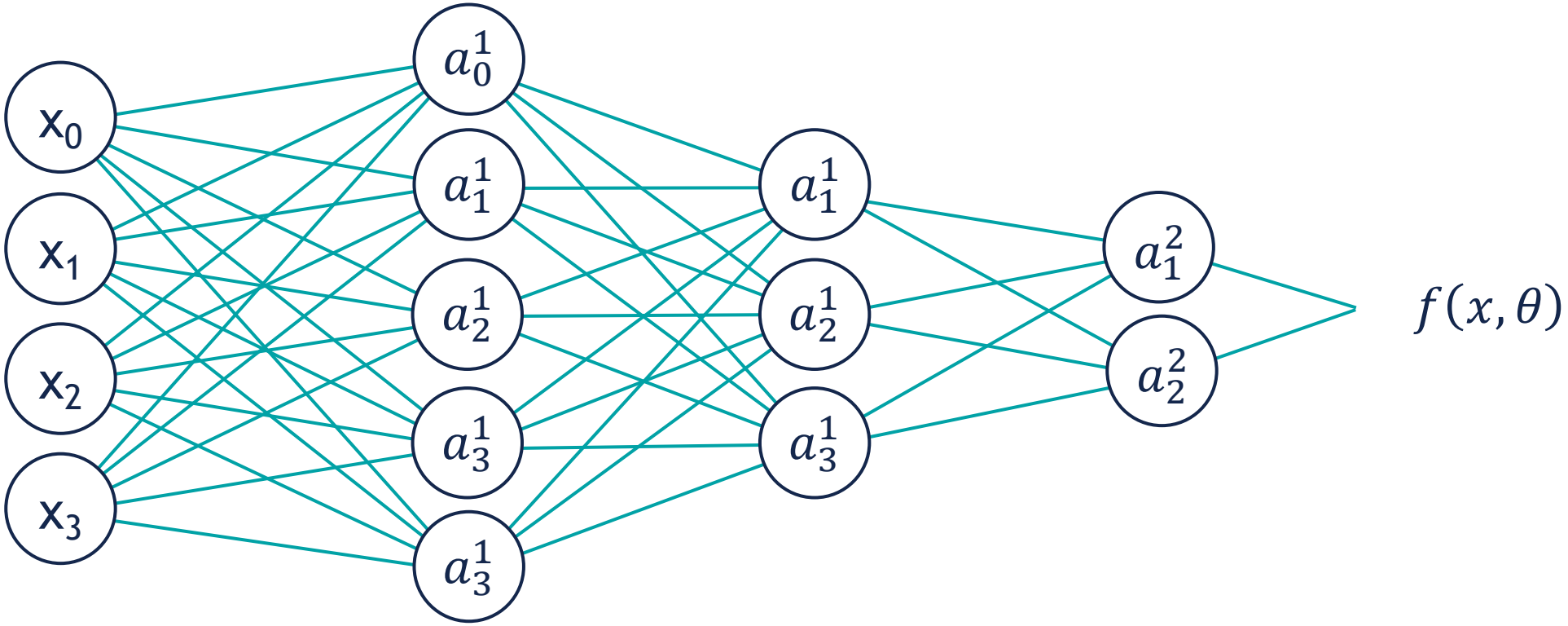
$$f(x,\theta) = a(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_m x_m)$$



- Linear combination
  → Linear regression
- (sigmoid) activation function
  → Logistic regression

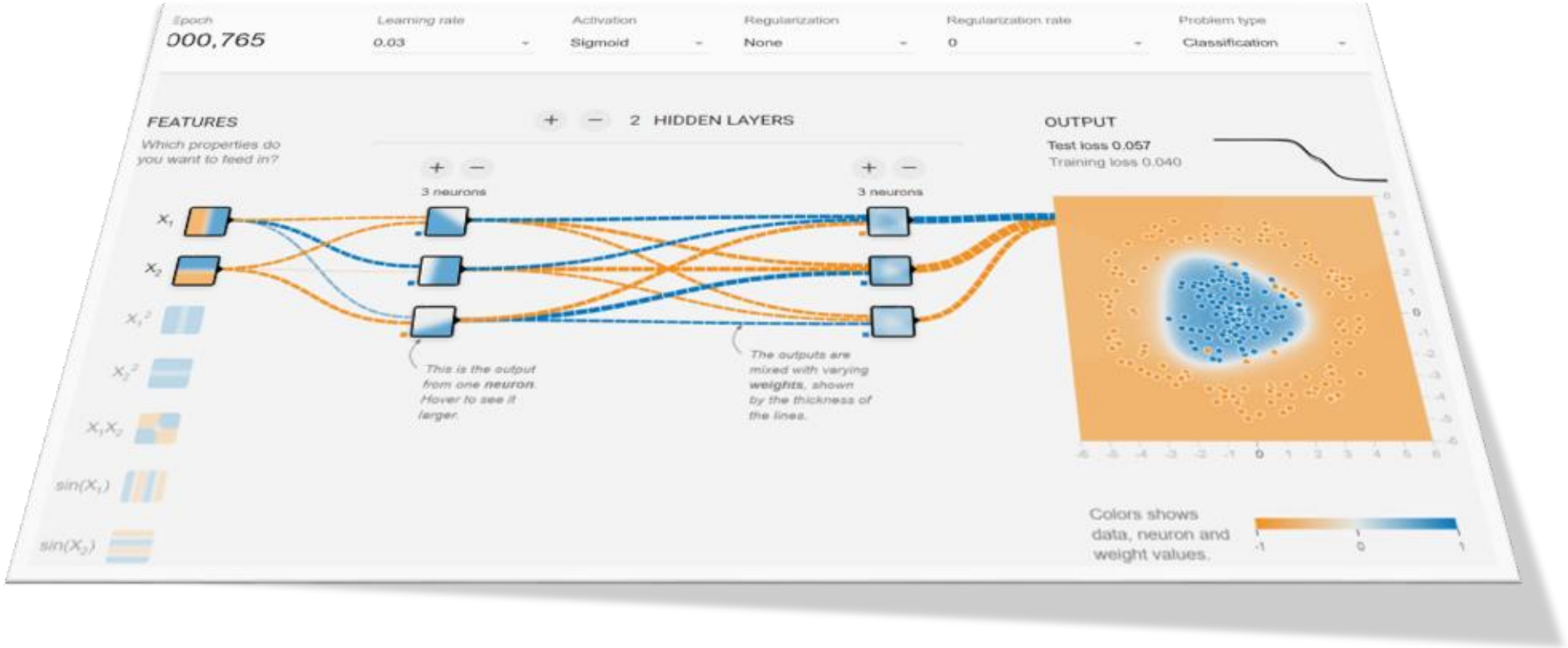
Neural network

= an architecture that consists of many perceptrons



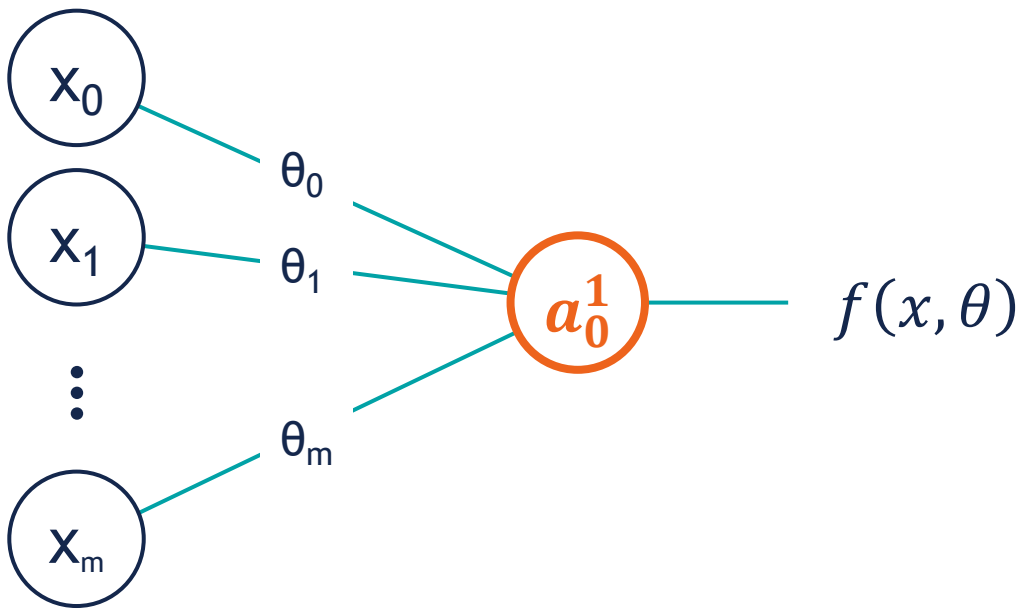
Neural network

Demo: create a small neural network



https://playground.tensorflow.org/

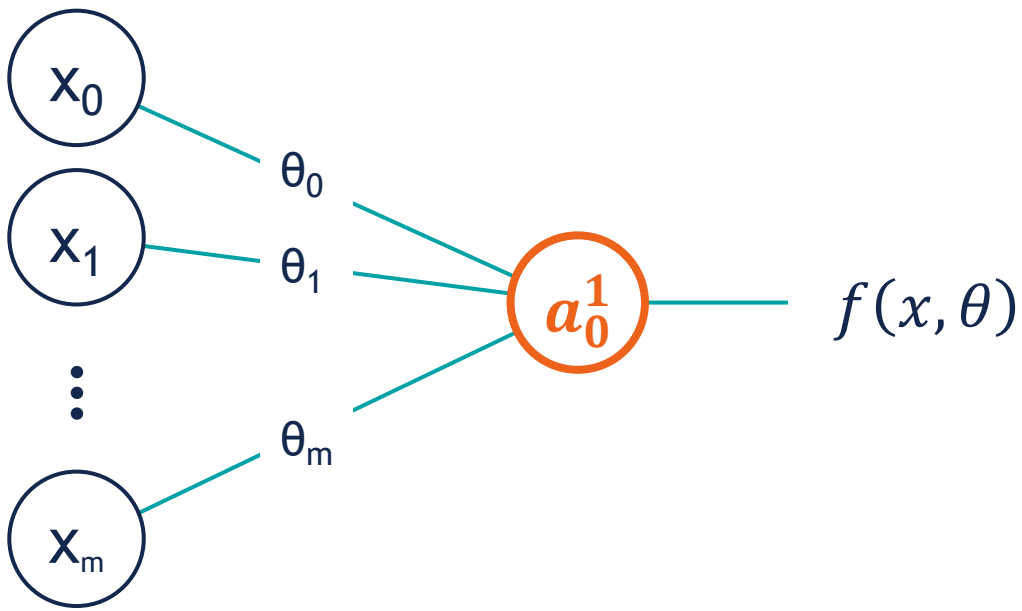
Activation functions



What happens if we use a linear activation function?

A combination of linear regressors = linear regressor

Activation functions

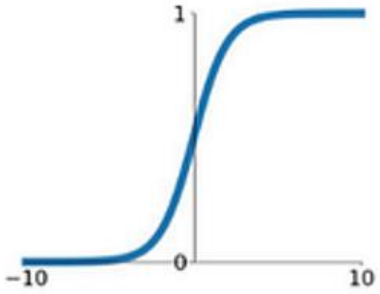


- Introduces non-linearity in the system
- Necessary to learn complex patterns in the data
- Popular functions: sigmoid, tanh, ReLU

Activation functions

Sigmoid

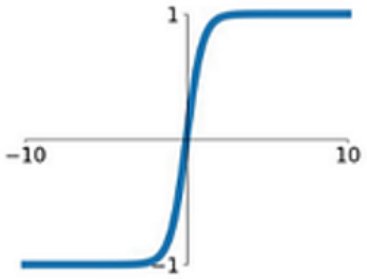
$$\sigma(x) = \frac{1}{1+e^{-x}}$$



"Mimics a neuron firing"

tanh

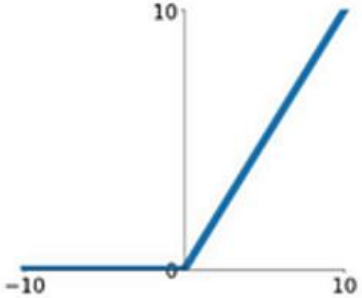
$$\tanh(x)$$



"Scaled sigmoid", zero-centered

ReLU

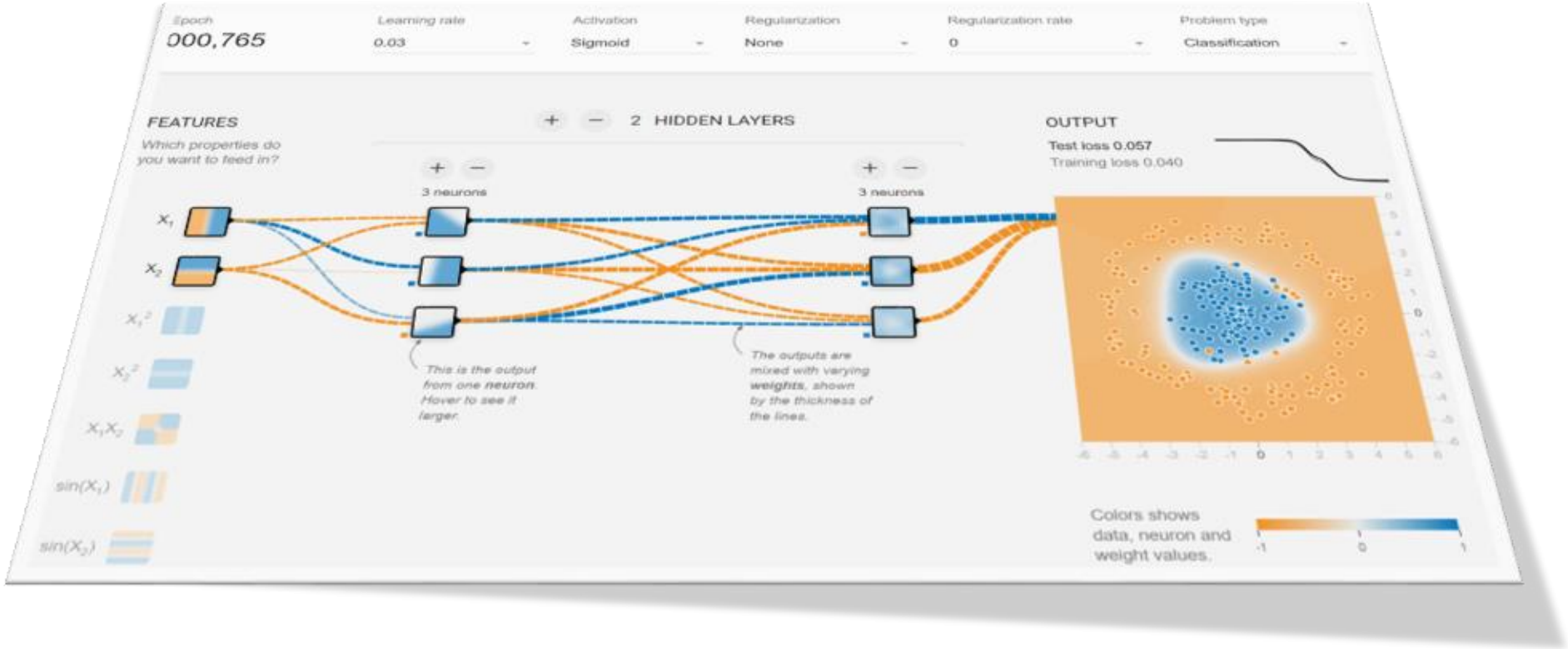
$$\max(0, x)$$



Simple to compute, but not zero-centered

Activation functions

Non-linear activation functions increase the expressiveness of the model



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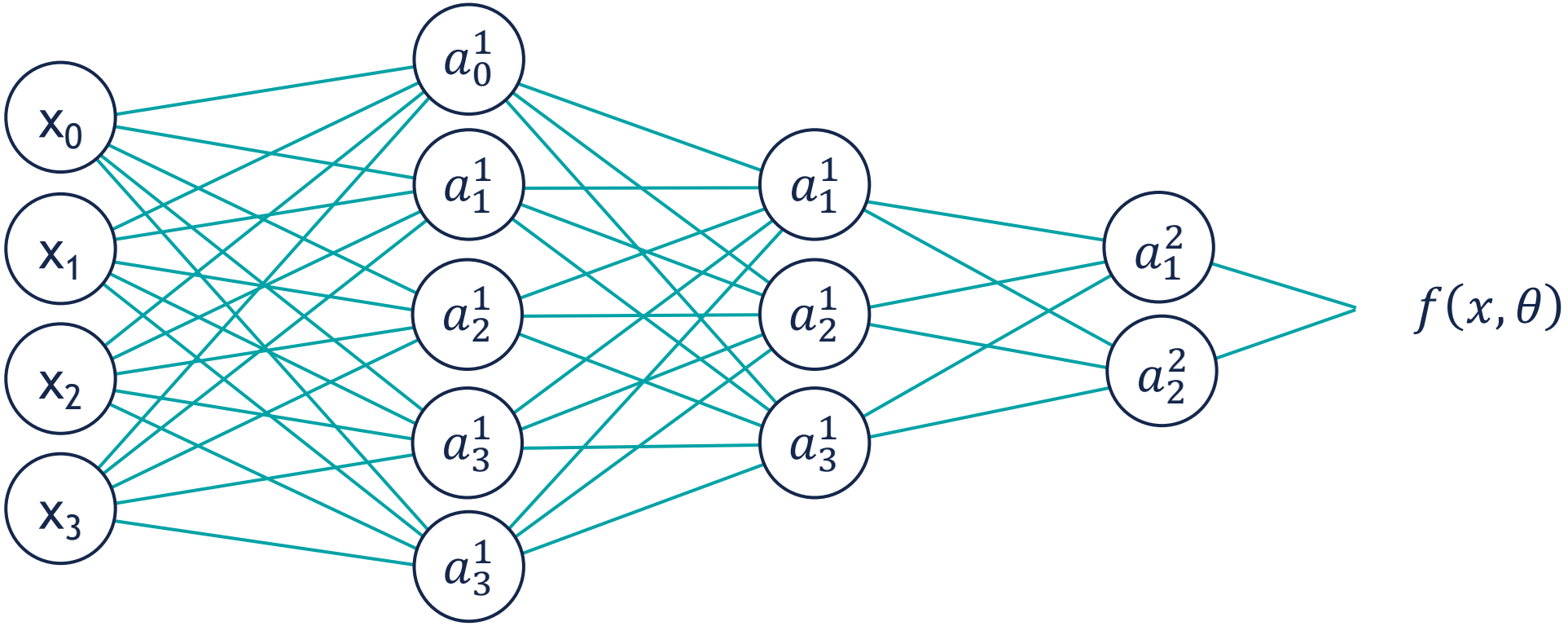
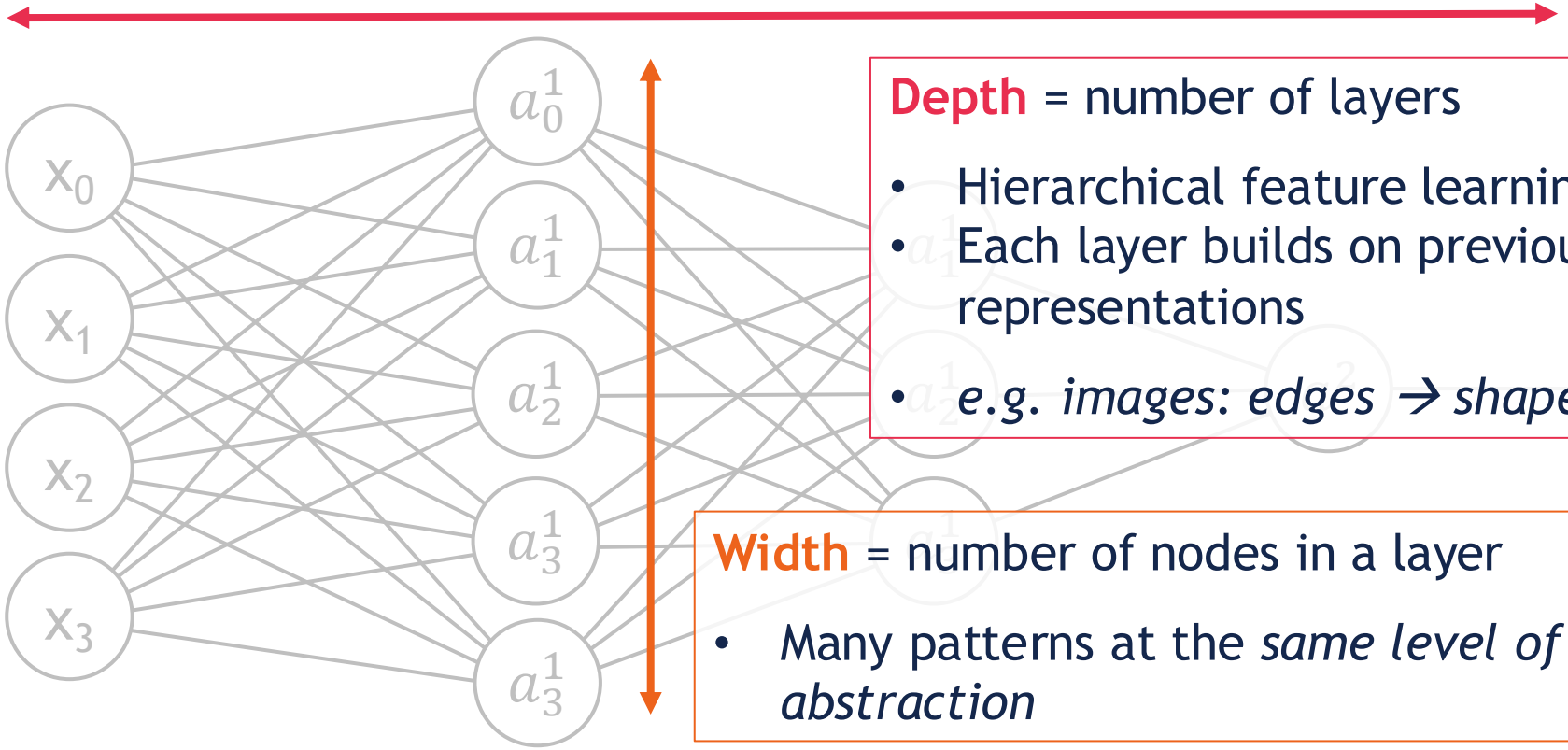


Input
(4 features)

Hidden layers
(5 and 3 nodes)

Output layer

Network architecture



Depth = number of layers

- Hierarchical feature learning
- Each layer builds on previous representations
- *e.g. images: edges → shapes → objects*

Width = number of nodes in a layer

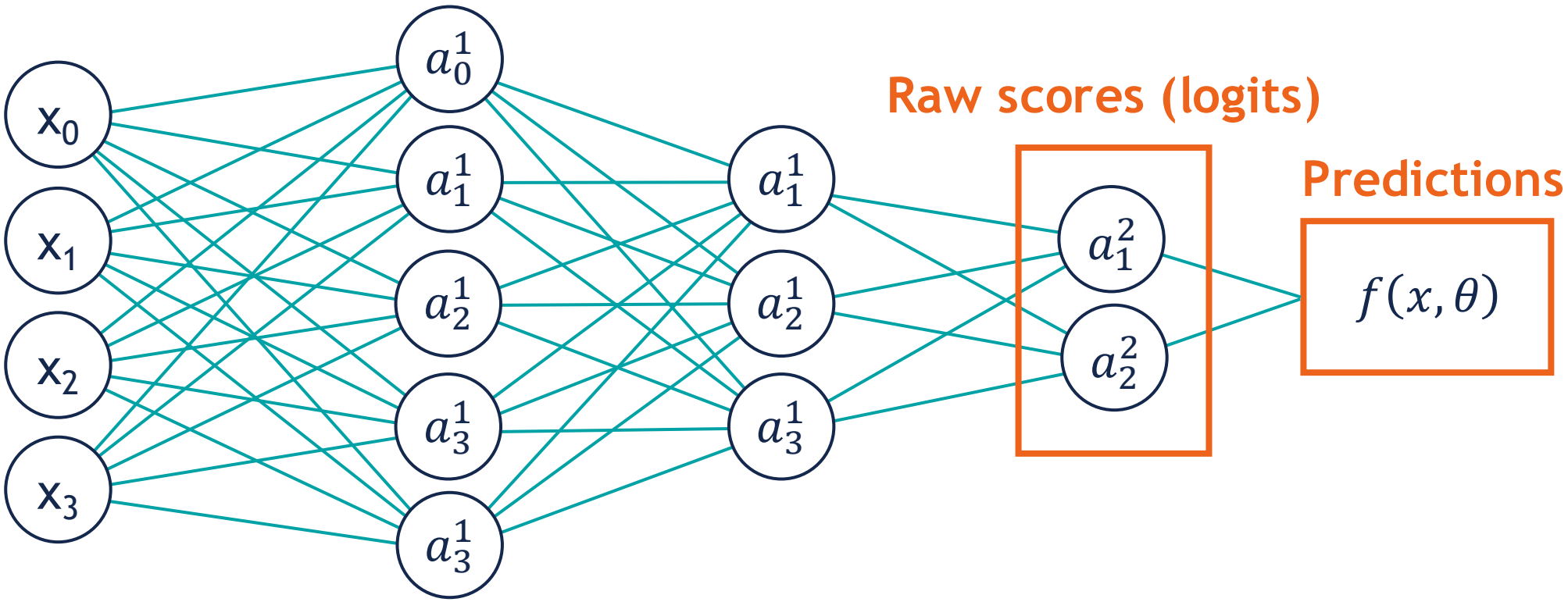
- Many patterns at the *same level of abstraction*

Input
(4 features)

Hidden layers
(5 and 3 nodes)

Output layer

Output layer



Input
(4 features)

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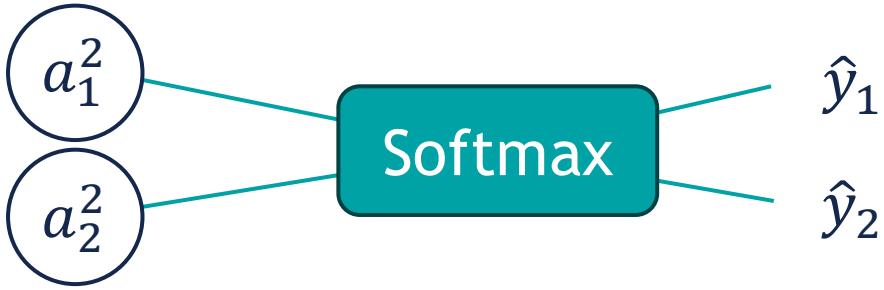
Output layer

VIB

Softmax

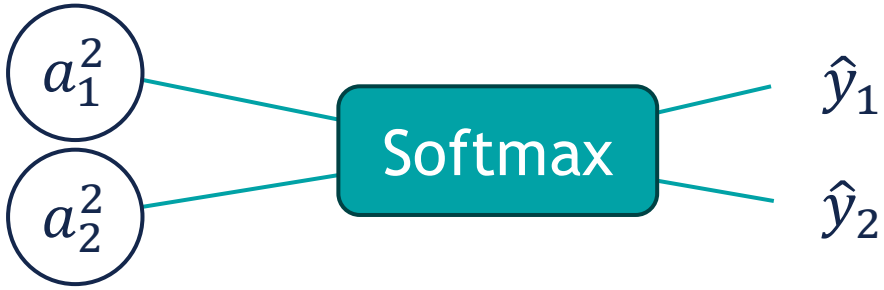


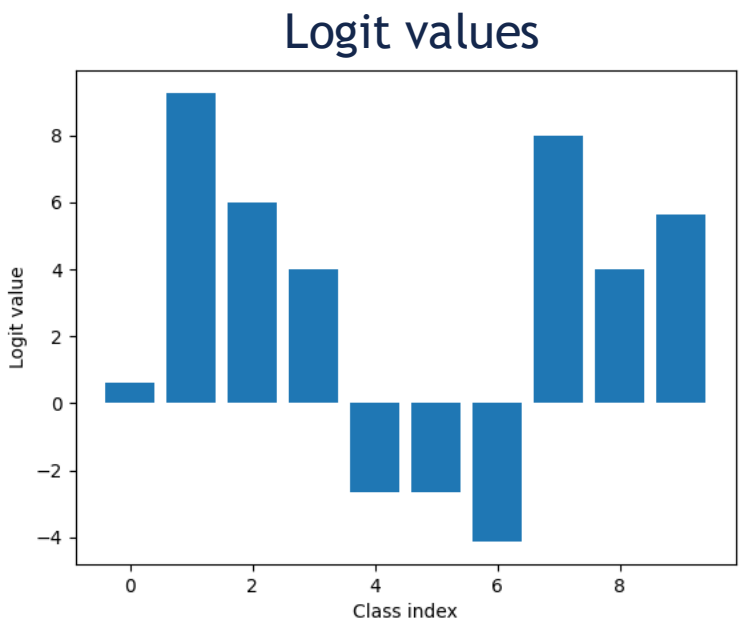
- Softmax function transforms logits to confidence scores

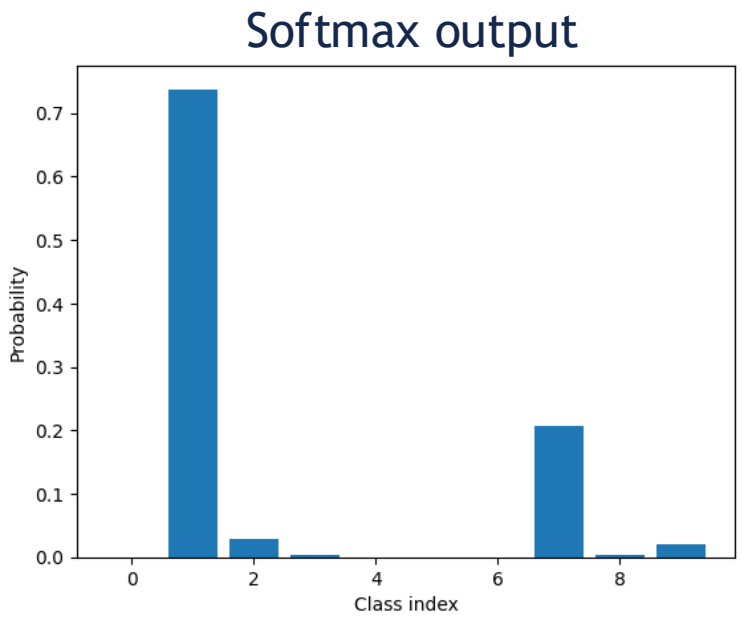
$$\sigma(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

- Exponentials exaggerate the differences
- Normalization produces probabilities

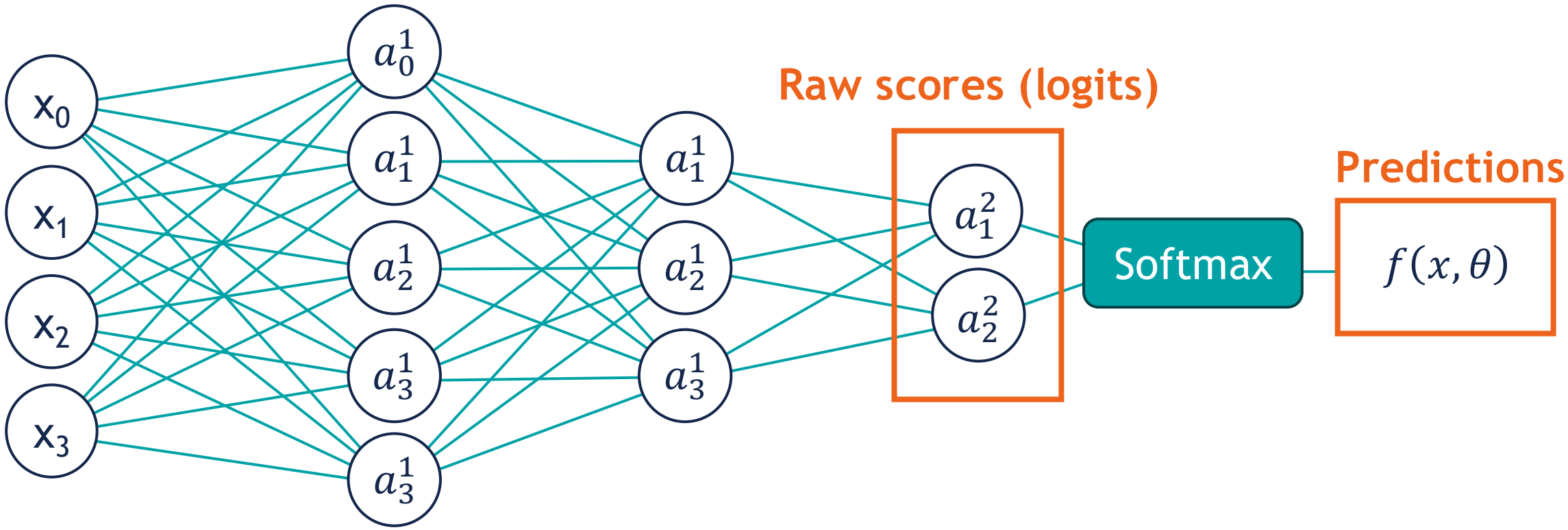
Softmax


a_1^2
a_2^2
Softmax
$\hat{y}_1$
$\hat{y}_2$


Logit values
Logit value
Class index
8
6
4
2
0
−2
−4
0
2
4
6
8


Softmax output
Probability
Class index
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
2
4
6
8

Output layer



Input
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Hidden layers
(5 and 3 nodes)

Output layer

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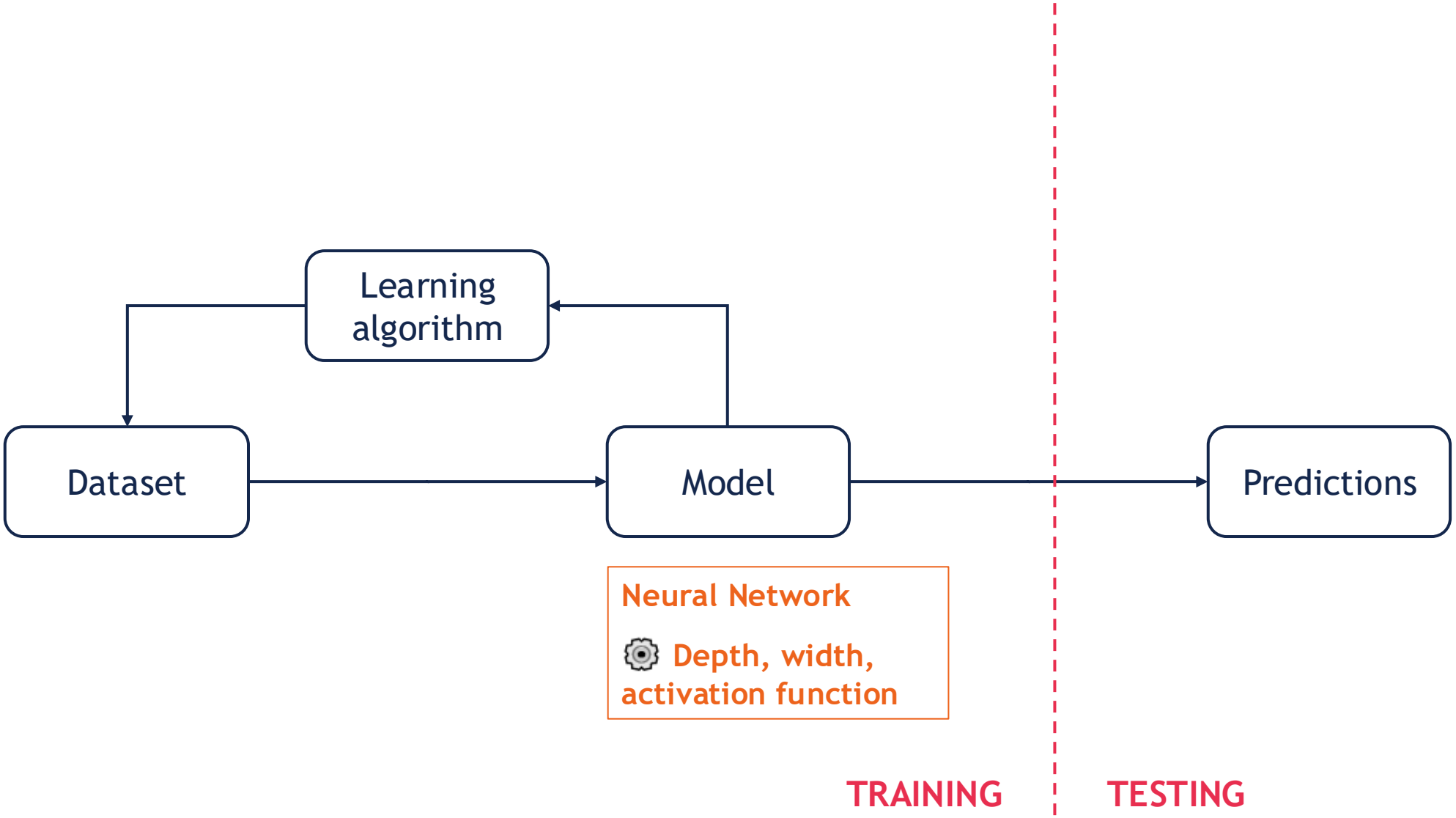
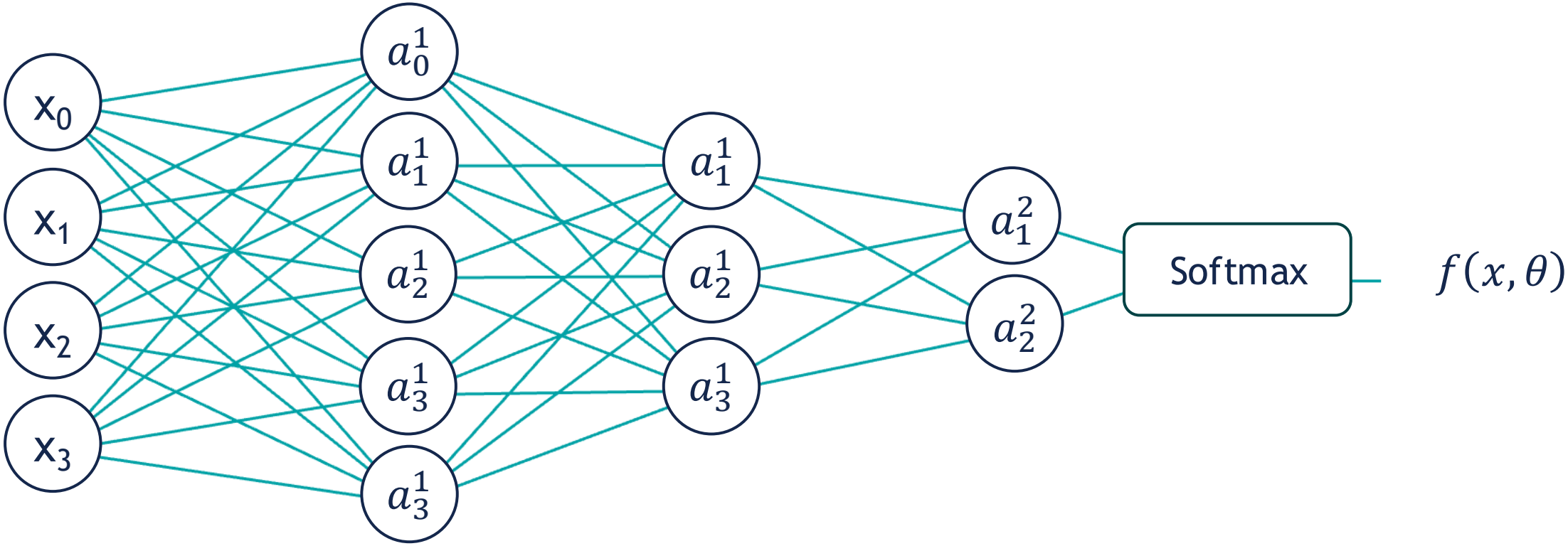


Forward pass

Compute the predictions based on the features (output from layer N is the input for layer N+1)



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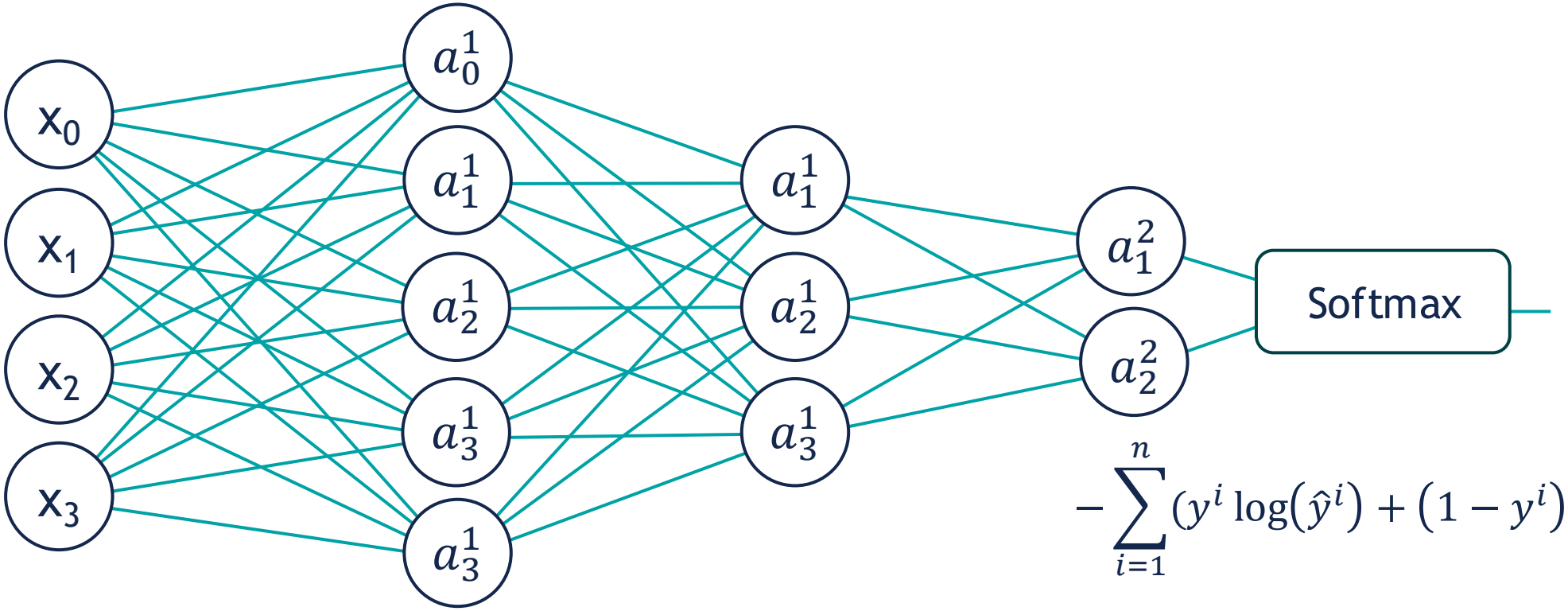
Loss function

Cross entropy quantifies how far the predictions are from the true values



$$-\sum_{i=1}^{n}\left(y^i \log(\hat{y}^i) + (1-y^i)\log(1-\hat{y}^i)\right)$$

Input
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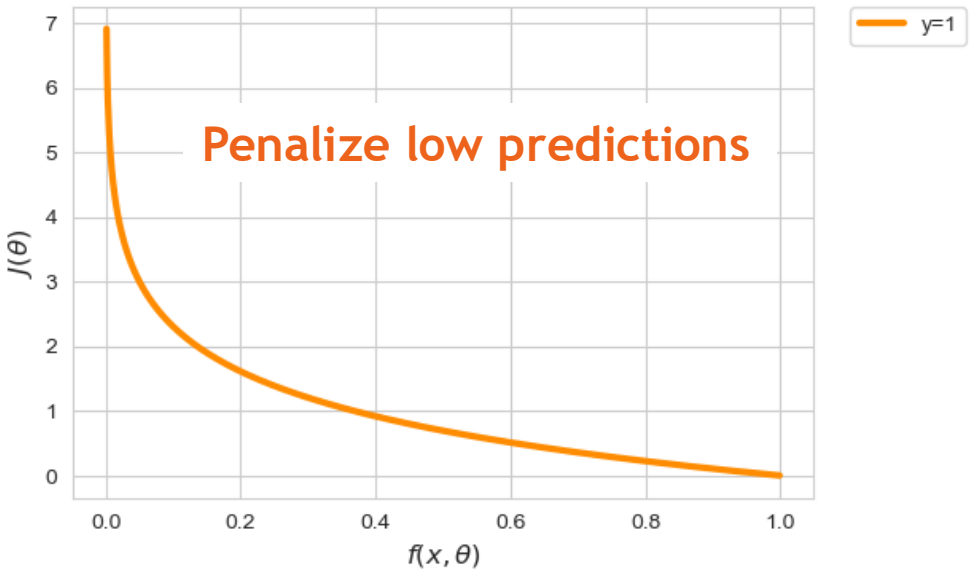
Output layer

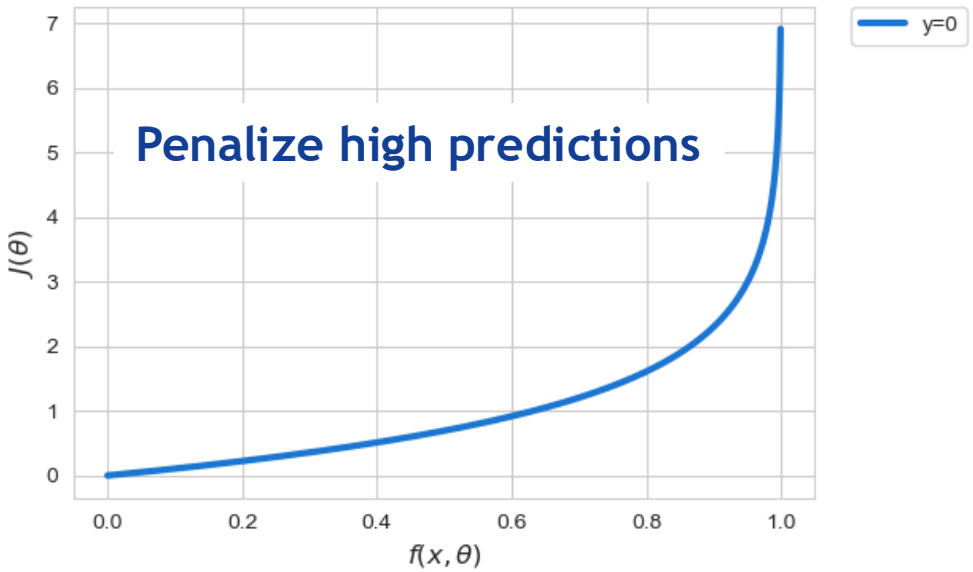
Logistic regression - fitting

$$J(\theta) = -[\frac{1}{n}\sum_{i=1}^{n} y^{(i)} \log\left(f\left(x^{(i)},\theta\right)\right) + (1-y^{(i)})\log(1-f\left(x^{(i)},\theta\right)$$

$y^{(i)} = 1$

$y^{(i)} = 0$

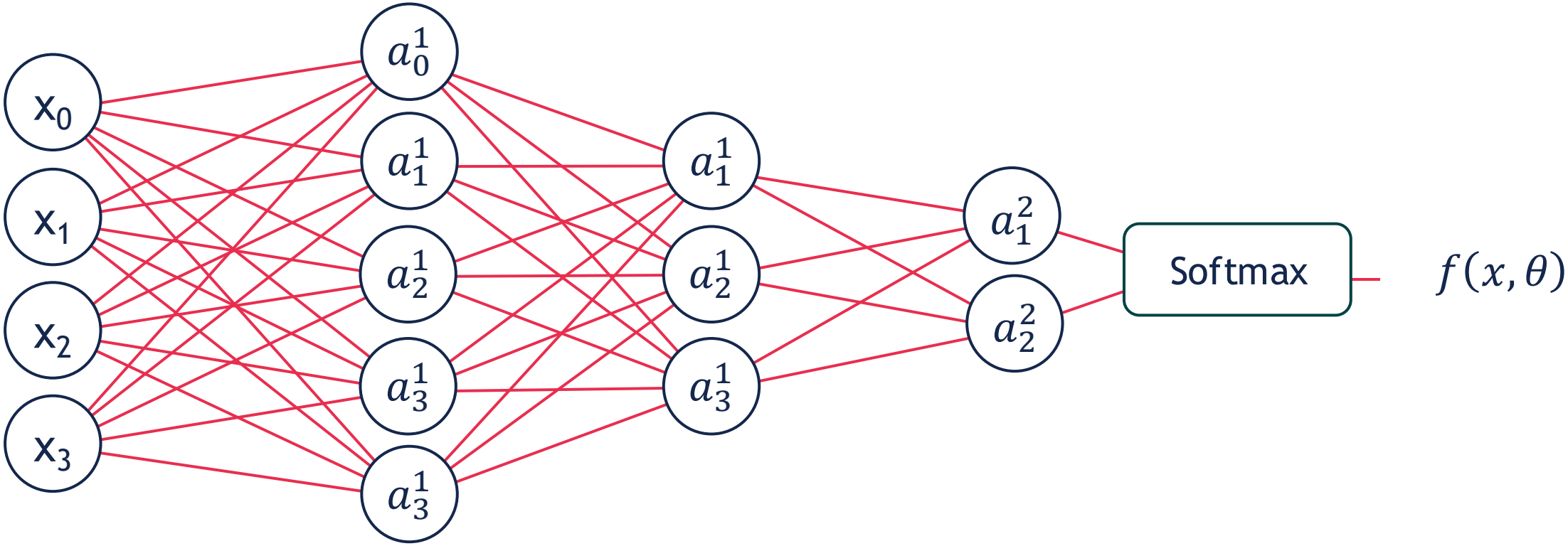




Backpropagation

Adjust the weights based on the prediction errors (i.e. gradient descent)

Chain rule → gradients can be computed per layer



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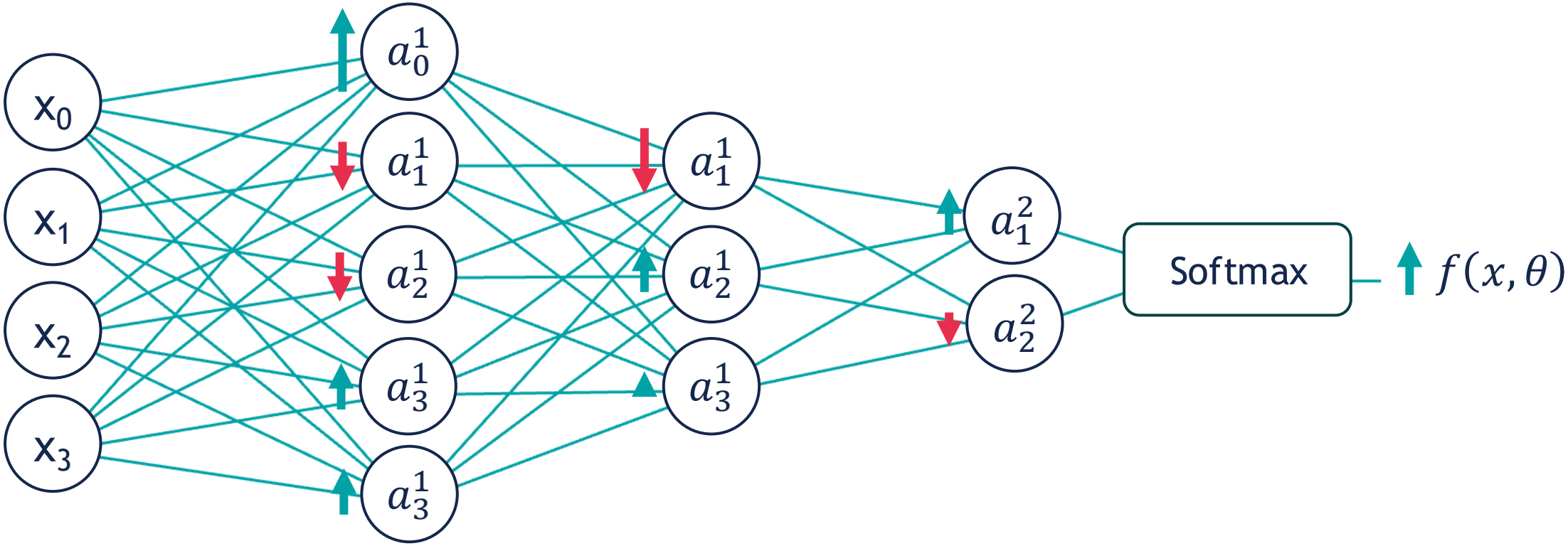

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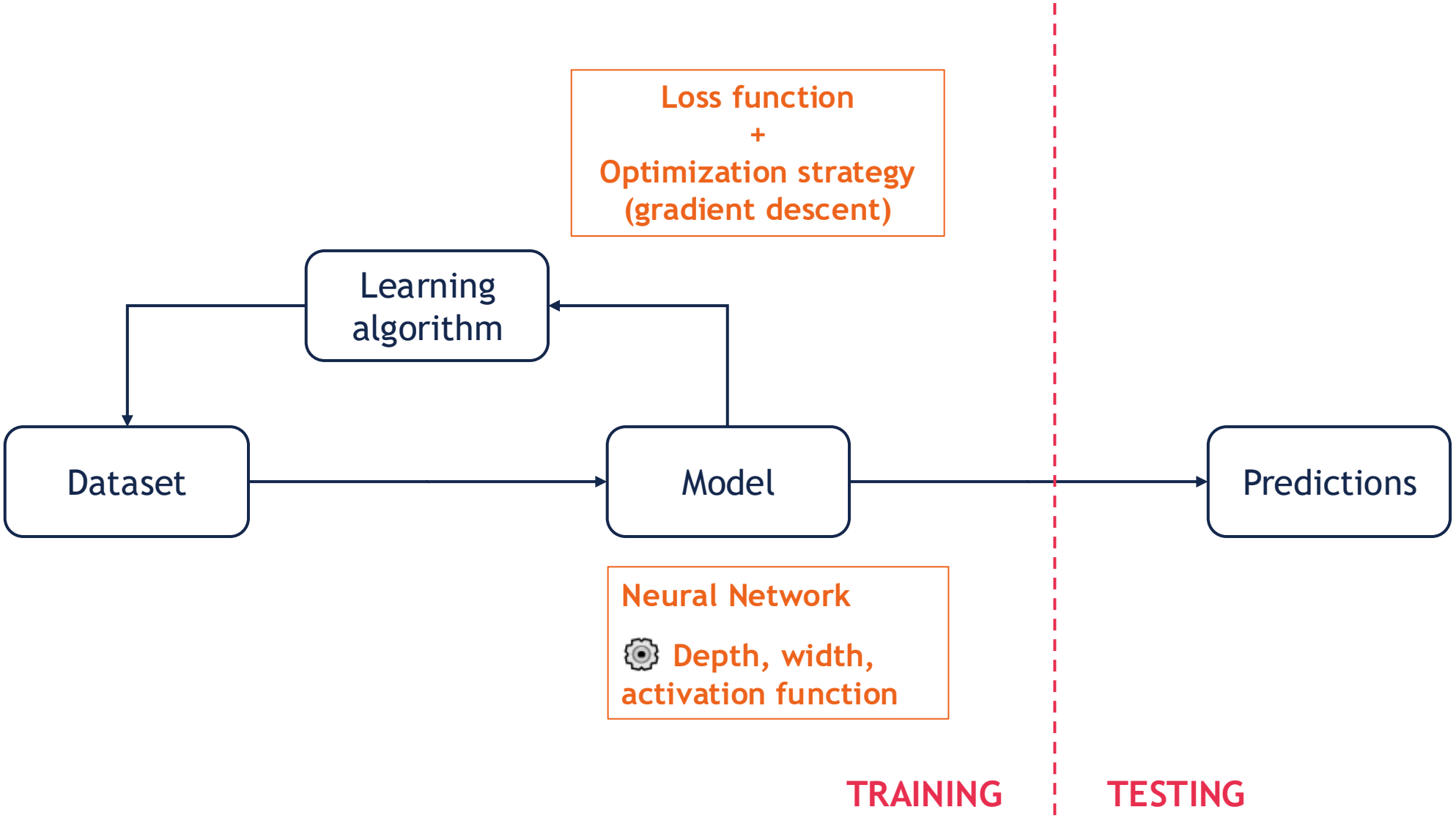
Training a neural network



VIB